Coil Sampling & Tracking Management System

Developed for Tata Steel Industry

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1. Project Overview

The Coil Sampling & Tracking Management System is a secure, web-based industrial application developed to digitally manage and monitor the complete coil sampling lifecycle at Tata Steel. The system integrates real-time tracking, KPI dashboards, reporting, logging, alarms, and secure user authentication into a centralized platform.

The application ensures operational transparency, traceability, process optimization, and enterprise-level data security.

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2. Objectives

- Digitize and automate the coil sampling workflow.
- Enable real-time tracking of coils across production stations.
- Manage Primary Data Input (PDI) and Primary Data Output (PDO).
- Provide KPI-driven dashboards for performance monitoring.
- Implement secure authentication using JWT-based authorization.
- Ensure audit traceability through logs and alarms.

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3. System Modules

A. Process Dashboard

- Real-time KPIs:
    - Sample Rate (Samples/hr)
    - Scrap Weight (Ton)
    - Number of Coils Sampled
    - Yield (%)
    - Average Time per Cycle
- Daily / Weekly / Monthly analytics.
- Scrap trend graphical visualization.

B. Primary Data Input (PDI)

Stores production-level coil details:

- Coil ID
- Steel Specification Code
- Thickness, Width, Length
- Weight
- Coil Inside & Outside Diameter
- Quality Code
- Production Date-Time

C. Primary Data Output (PDO)

Stores processed sampling data:

- TDC Number
- Grade Description
- Sample Cut Date-Time
- Downloadable structured reports

D. Coil Tracking System

Visual real-time station tracking:

- Entry Station 1 & 2
- Band Cutter Station
- Sampling Station
- Circumferential Banding Station
- Weighing Station
- Delivery Station

Features:

- Direction-of-flow visualization
- Live coil presence indicator
- Station-wise status monitoring

E. Logs Management

1. Telegram Log

- Tracks TCP/IP communication
- Send/Receive direction

- Outer system identification
- Telegram number & description
- Communication status

2. Activity Log

- Login tracking
- OTP generation & verification
- Password change events
- Device login tracking
- IP address logging for auditing

F. Alarm Management

- Real-time alarm generation
- Severity classification
- Parameter identification
- Timestamped event history

G. Reports Module

- Custom date-range filtering
- Feature-based filtering
- Parameter selection
- Downloadable report generation
- SQL-based optimized querying

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4. Security Architecture & Authentication

JWT-Based Secure Authentication

The system implements **JSON Web Token (JWT)** based authentication to ensure secure and stateless user sessions.

Authentication Flow:

1. User Login with credentials
2. Password verified using secure hashing
3. Server generates signed JWT token
4. Token sent to frontend
5. Token stored securely (HTTP-only cookie / secure storage)
6. Token validated on each API request

Security Features Implemented

- JWT Token signed using **HMAC SHA-256**
- Token expiration & refresh mechanism
- Role-Based Access Control (RBAC)
- ASP.NET Authorization Middleware
- Password hashing using salted hashing algorithm (e.g., PBKDF2 / BCrypt)
- HTTPS secure communication
- SQL Injection prevention using parameterized queries
- Input validation & model validation
- IP logging & device tracking
- OTP-based secondary verification
- Secure API endpoints with `[Authorize]` attributes

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5. Technology Stack

Layer	Technology Used
Frontend	Angular
Backend	ASP.NET (.NET Core)
Database	Microsoft SQL Server
Authentication	JWT (JSON Web Token)
Communication	TCP/IP
Architecture	3-Tier Architecture

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6. Key Features

- Industrial real-time dashboard
- Secure JWT authentication
- OTP-based login verification
- Role-based authorization
- Full audit logging
- Real-time coil station tracking
- Alarm severity monitoring
- Downloadable reports
- Structured SQL database design

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7. Impact & Benefits

- Reduced manual dependency
- Improved production traceability

- Enhanced security compliance
- Better operational decision-making
- Real-time monitoring of sampling workflow
- Secure enterprise-grade authentication

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8. Conclusion

The Coil Sampling & Tracking Management System successfully integrates production monitoring, sampling management, reporting, and secure authentication into a single enterprise-grade platform.

By combining Angular frontend, ASP.NET backend, SQL Server database, JWT-based authentication, and real-time KPI dashboards, the system enhances operational efficiency, data security, and process transparency within Tata Steel's industrial workflow.